JIWON KIM

Ph.D. Candidate in Computer Science at Purdue University
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250 N University St, West Lafayette, Indiana 47907

Research Interests

Security for programmable networks, such as Software-Defined Networking (SDN), Intent-Based Networking (IBN), and programmable data planes with P4

Education

Aug 2020 – *May 2027* (Expected)	**Purdue University**, West Lafayette, IN Ph.D. in Computer Science Major: *Programmable Network Security* | Advisor: Prof. Dave (Jing) Tian
Mar 2015 – Aug 2016	**Sungkyunkwan University**, Suwon, Republic of Korea Master of Science in Platform Software Major: *System Software for Non-Volatile Memory* | Advisor: Prof. Hwansoo Han
Mar 2011 – Feb 2015	**Sungkyunkwan University**, Suwon, Republic of Korea Bachelor of Science in Software Engineering

Research and Professional Experience

Research Assistant
Aug 2020 – *Current*

PurSec Lab, Purdue University, West Lafayette, IN
Advisor: Prof. Dave (Jing) Tian
- Led research on fuzzing P4 networks [P2]
- Led research on fuzzing intent-based networking [P5]
- Participated in research projects on securing intent-based networking [P1,P3,P4]

Lead Grad Student
Aug 2025 – May 2026
Aug 2024 – May 2026

Purdue University, West Lafayette, IN
PurSec Lab | 4 PIs: Prof. Antonio Bianchi, Prof. Z. Berkay Celik, Prof. Dave (Jing) Tian, Prof. Dongyan Xu
Prof. Dave (Jing) Tian's Research Group
- Organized weekly security reading groups (10-20 attendees) spanning four PI groups
- Coordinated lad-wide seminars and social events for 20-student group

SWE Intern
May 2026 – Aug 2026
May 2025 – Aug 2025

Meta (NetSec), Menlo Park, CA
- Designed formal network policy analysis with ILP heuristics and SMT solvers, reducing time from hours to seconds
- Designed provider-agnostic cloud network visualization tool and implemented the AWS instantiation

Research Engineer
(Full-Time)
Sep 2016 – Jul 2020

ATTO Research, Seoul, Republic of Korea
- Created flow-specific database for concurrent and consistent software-defined network updates [K1,K2]
- Designed SDN applications to manage underlay/overlay networks in private cloud [K3]
- Developed secure SDN controller cluster with TLS, RBAC, and flow monitoring [K4–9]
- Implemented domain-specific language and command-line interface for SDN controller

Research Assistant
Mar 2014 – Aug 2016

ARCS Lab, Sungkyunkwan University, Suwon, Republic of Korea
Advisor: Prof. Hwansoo Han
- Participated in research projects on system software for non-volatile memory [P6,P7][U1,K10]

Research Intern
Aug 2013 – Feb 2014

M2M Lab, Purdue University, West Lafayette, IN
NIPA-Purdue Capstone Project - Advisor: Prof. Eric Matson
- Designed road guidance system for the visually impaired

Intern
Jan 2013 – Feb 2013

Acryl Inc., Seoul, Republic of Korea
- Developed prototype of Android messenger application

Intern
Feb 2012

Hancom Corp., Seongnam, Republic of Korea
- Participated in quality assurance team

Publications

[P1] **"Exploiting Intent-Flow State Vulnerabilities in Intent-Based Networking"**
Angela Yan, **Jiwon Kim**, Benjamin E. Ujcich, Dave (Jing) Tian.
IEEE Secure Development Conference (**SecDev '25**), Oct 2025.

[P2] **"Chimera: Fuzzing P4 Network Infrastructure for Multi-Plane Bug Detection and Vulnerability Discovery"**
Jiwon Kim, Dave (Jing) Tian, Benjamin E. Ujcich.
46th IEEE Symposium on Security and Privacy (**S&P '25**), May 2025.
(acceptance rate: 257/1740 = 14.8%)

[P3] **"Exploiting Temporal Vulnerabilities for Unauthorized Access in Intent-Based Networking"**
Ben Weintraub, **Jiwon Kim**, Ran Tao, Cristina Nita-Rotaru, Hamed Okhravi, Dave (Jing) Tian, Benjamin E. Ujcich.
31st ACM Conference on Computer and Communications Security (**CCS '24**), Oct 2024.
(acceptance rate: 331/1964 = 16.9%)

[P4] **"Security Challenges of Intent-Based Networking"**
Jiwon Kim, Dave (Jing) Tian, Hamed Okhravi, Benjamin E. Ujcich.
Communications of the ACM (**CACM**), vol. 67(7), pp. 56-65, Jul 2024.

[P5] **"Intender: Fuzzing Intent-Based Networking with Intent-State Transition Guidance"**
Jiwon Kim, Benjamin E. Ujcich, Dave (Jing) Tian.
32nd USENIX Security Symposium (**Security '23**), Aug 2023.
(acceptance rate: 422/1444 = 29.2%[1]))

[P6] **"Efficient Memory Mapped File I/O for In-Memory File Systems"**
Jungsik Choi, **Jiwon Kim**, Hwansoo Han.
9th USENIX Workshop on Hot Topics in Storage and File Systems (**HotStorage '17**), Jul 2017.

[P7] **"In-Memory File System with Efficient Swap Support for Mobile Smart Devices"**
Jungsik Choi, Joonwook Ahn, **Jiwon Kim**, Sungtae Ryu, Hwansoo Han.
IEEE Transactions on Consumer Electronics, vol. 62(3), pp. 275-282, Aug 2016.

Academic Services

Organizing Committee

- IEEE Secure Development Conference (**SecDev**): 2025 (Registration Chair)

Technical Program Committee

- USENIX Security Symposium (**Security**): 2027
- ACM Conference on Data and Application Security and Privacy (**CODASPY**): 2027
- Hardware and Architectural Support for Security and Privacy (**HASP**): 2026
- USENIX WOOT Conference on Offensive Technologies (**WOOT**): 2026
- EAI International Conference on Security and Privacy in Cyber-Physical Systems and Smart Vehicles (**SmartSP**): 2026
- IEEE Workshop on the Internet of Safe Things (**SafeThings**): 2023, 2025

Journal Reviewer

- IEEE Transactions on Dependable and Secure Computing (**TDSC**): 2026

Artifact Evaluation Committee

- USENIX Security Symposium (**Security**): 2025, 2026 – *Noteworthy Reviewer Recognition (2026)*

External Reviewer

- ACM ASIA Conference on Computer and Communications Security (**ASIACCS**): 2024
- Network and Distributed System Security Symposium (**NDSS**): 2024
- International Symposium on Research in Attacks, Intrusions, and Defenses (**RAID**): 2023
- Future Generation Computer Systems: 2026

[1] In the 2023 cycle, USENIX Security modified its revision process, leading to an unusually high acceptance rate relative to the historical range of 15.5%–19.1% observed between 2013 and 2022.

Mentoring

Fall 2026 – *Current*	**Colin Crist**, MS, Purdue University
Fall 2024 – Fall 2025	**Angela Yan**, Undergraduate Researcher, Purdue University Project: Finding vulnerabilities in intent-based networking controller [P1]

Talks

- Uncovering Security Vulnerabilities in Programmable Networks, Georgetown University, Oct 2026
- Fuzzing P4 Network Infrastructure for Multi-Plane Bug Detection and Vulnerability Discovery, Institute for Networking and Security Research (INSR), Penn State University, Mar 2026
- Securing Intent-Based Networking, Lightning Talk, Midwest Security Workshop, Indiana University Bloomington, Sep 2025
- Finding Semantic Bugs in Intent-Based Networking with Fuzzing, Sungkyunkwan University, Dec 2023

Awards

Aug 2020 – Aug 2024	**Ross Fellowship**, Purdue University
Aug 2023	**College of Science Graduate Student Travel Grant**, Purdue University
Aug 2018	**Hall of Fame**, ATTO Research
Mar 2016 – Aug 2016	**Simsan Scholarship**, Sungkyunkwan University
Dec 2015	**Outstanding Paper Award**, Korea Institute of Information Scientists and Engineers
Mar 2015 – Aug 2016	**Masters and Doctors Connected Track Scholarship**, Sungkyunkwan University
Oct 2012, Apr 2013	**Dean's List**, Sungkyunkwan University
Mar 2011 – Feb 2015	**National Science & Technology Scholarship**, Korea Student Aid Foundation

Teaching Experience

Guest Lecture

Spring 2024	"Intent-Based Network Security," **CS 528 – Network Security**, Purdue University

Teaching Assistant

Spring 2026	**CS 536 – Data Communication and Computer Networks**, Purdue University
Spring 2016	**SWE 2013 – Software Creative Project**, Sungkyunkwan University
Fall 2015	**SWE 3010 – Compiler Design**, Sungkyunkwan University
Fall 2015	**ICE 2002 – Data Structures**, Sungkyunkwan University
Spring 2015	**GEDB 029 – Basis and Practice in Programming**, Sungkyunkwan University

Reported Vulnerabilities and CVEs

CVE-2024-53423	P4 pipeline interpreter DoS vulnerability, ONOS
CVE-2024-24270	disclosure pending
CVE-2023-45477	disclosure pending
CVE-2023-43889	disclosure pending
CVE-2023-43888	disclosure pending
CVE-2023-41591	P4 host registration vulnerability, HostManager, ONOS
CVE-2022-29944	Mishandled intent conflict vulnerability, IntentManager, ONOS
CVE-2022-29609	Misleading intent state vulnerability, IntentManager, ONOS
CVE-2022-29608	Traffic loop vulnerability, IntentManager, ONOS
CVE-2022-29607	Misleading intent state vulnerability, IntentManager, ONOS
CVE-2022-29606	Switch port range vulnerability, IntentManager, ONOS
CVE-2022-29605	Switch IP version inconsistency vulnerability, IntentManager, ONOS
CVE-2022-29604	Switch ID inconsistency vulnerability, IntentManager, ONOS
CVE-2022-24109	Duplicate intent removal vulnerability, IntentManager, ONOS
CVE-2022-24035	Intent purge operation vulnerability, IntentManager, ONOS
CVE-2021-38364	Intent manipulation vulnerability, IntentManager, ONOS
CVE-2021-38363	pendingMap vulnerability, IntentManager, ONOS

Patents

US Patents

[U1] "Memory mapping management method for nonvolatile main memory system"
Jungsik Choi, Hwansoo Han, **Jiwon Kim**.
US Patent 10,185,654, 2019.

Korean Patents

[K1] "Method, apparatus and computer program for processing flow rule transactions in software defined network"
Jiwon Kim, Jaekyun Shim, Hyunwoo Park, Yongjoo Song.
KR 10-2275765, 2021.

[K2] "Method, apparatus and computer program for operating the flow rules database in software defined network"
Jiwon Kim, Jaekyun Shim, Yongjoo Song, Jonghyun Kwon.
KR 10-2226915, 2021.

[K3] "Method, apparatus and computer program for supporting distributed snat in a cloud environment in a software defined network"
Jiwon Kim, Yongjoo Song.
KR 10-2200402, 2021.

[K4] "Methods, apparatus and computer programs for providing high availability and reliability in software defined network"
Jiwon Kim, Jaekyun Shim, Yongjoo Song.
KR 10-2092091, 2020.

[K5] "Method, apparatus and computer program for tracking traffic using software defined networking"
Yongjoo Song, **Jiwon Kim**.
KR 10-2066555, 2020.

[K6] "Method and computer program for handling trouble using packet-out message in software defined networking environment"
Hyunwoo Park, **Jiwon Kim**, Inwook Hwang, Jaeho Kim, Beomsoo Park, Yongjoo Song.
KR 10-1969304, 2019.

[K7] "Method, apparatus and computer program for defending software defined network"
Jiwon Kim, Hyunwoo Park, Inwook Hwang, Jaewoong Chung.
KR 10-1932656, 2018.

[K8] "Method and computer program for handling trouble using flow-mod message in software defined networking environment"
Hyunwoo Park, **Jiwon Kim**, Inwook Hwang, Jaeho Kim, Beomsoo Park, Yongjoo Song.
KR 10-1931543, 2018.

[K9] "Method, apparatus, and computer program for verifying host status information in a software defined network"
Jiwon Kim, Inwook Hwang, Yongjoo Song.
KR 10-1931139, 2018.

[K10] "Method for atomic update of memory mapped files stored in non-volatile memory and control apparatus thereof"
Hwansoo Han, Jungsik Choi, **Jiwon Kim**.
KR 10-1881038, 2018.

Last updated on November 2, 2026